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Energy Forecasting Using RNN, LSTM, GRU, and CNN–RNN

Performance Evaluation and Comparative Analysis — Answers to Questions 1–6

Evidence Base: Fully Executed Google Colab Notebook (25/25 experiments; NVIDIA Tesla T4 GPU). All metrics are measured test-set outputs, not simulated results.

1. Modify the Provided RNN, LSTM, and GRU Notebooks

An additional same-type recurrent layer containing 32 neurons was added to every original sequence architecture. In the One-to-One, Many-to-One, and synchronized Many-to-Many variants, the modified models have two 32-unit recurrent layers. In One-to-Many and Seq2Seq, the new 32-unit layer was added to the existing encoder–decoder architecture while retaining the decoder, yielding three recurrent blocks in total.

Representative Single-Output Layer Changes

RNN: SimpleRNN (32, return_sequences=True) → SimpleRNN(32) → Dense(1)

LSTM: LSTM (32, return_sequences=True) → LSTM(32) → Dense(1)

GRU: GRU (32, return_sequences=True) → GRU(32) → Dense(1)

For synchronized Many-to-Many, the last recurrent layer returns a sequence and the prediction head is TimeDistributed(Dense(1)). For One-to-Many and Seq2Seq, the encoder output is passed through RepeatVector(6) and the retained recurrent decoder before the output layer.

Shared Experiment Configuration

Parameter	Configuration
Source dataset	Spanish hourly energy dataset (energy_dataset.csv)
Dataset size	35,064 hourly observations
Chronological split	24,544 training / 5,260 validation / 5,260 test
Features	Load actual; solar, wind and fossil gas generation; price actual
Prediction target	price actual (EUR/MWh)
Sequence settings	24-step input windows where applicable; 6-step forecast horizon for sequence forecasts
Training	Adam optimizer; MSE loss; batch size 64; 100 epochs
Preprocessing	Min-Max scaler fitted on training partition; test predictions inverse transformed
Hardware	Google Colab NVIDIA Tesla T4 GPU

A total of 25 models were executed: 15 stacked RNN/LSTM/GRU models, 5 original RNN baselines, and 5 CNN–RNN variants.

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2. Obtain MAE, MSE, RMSE, and R^2 for each Model Typology

All four metrics were calculated using the held-out test data after 100 epochs. Lower MAE/MSE/RMSE is preferable; higher R^2 is preferable. MAE and RMSE are in EUR/MWh, MSE is in (EUR/MWh)².

RNN — Stacked Recurrent Architecture

Typology	MAE	MSE	RMSE	R^2
One-to-One	2.1286	8.3684	2.8928	0.8660
Many-to-One	**1.5200**	**4.3343**	**2.0819**	**0.9308**
One-to-Many	5.3367	49.7112	7.0506	0.2049
Many-to-Many (Synced)	1.8877	6.0439	2.4584	0.9035
Many-to-Many (Seq2Seq)	3.4536	19.5835	4.4253	0.6876

LSTM — Stacked Recurrent Architecture

Typology	MAE	MSE	RMSE	R^2
One-to-One	2.0278	8.0564	2.8384	0.8710
Many-to-One	**1.5585**	**4.4213**	**2.1027**	**0.9294**
One-to-Many	6.2753	59.9712	7.7441	0.0408
Many-to-Many (Synced)	1.6112	4.7765	2.1855	0.9238
Many-to-Many (Seq2Seq)	3.2323	18.0725	4.2512	0.7117

GRU — Stacked Recurrent Architecture

Typology	MAE	MSE	RMSE	R^2
One-to-One	2.0549	8.0935	2.8449	0.8704
Many-to-One	**1.5267**	**4.2160**	**2.0533**	**0.9327**
One-to-Many	6.3144	60.4158	7.7728	0.0337
Many-to-Many (Synced)	1.6321	4.8465	2.2015	0.9227
Many-to-Many (Seq2Seq)	3.6996	23.5400	4.8518	0.6245

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3. Determine the Best-Performing Typology after Adding the Recurrent Layer

Using minimum test RMSE as the primary selection criterion and R^2 as supporting evidence, Many-to-One was the best-performing typology within all three recurrent families. It takes 24 hours of past feature measurements and predicts the next single electricity price.

Model	Selected typology	MAE	MSE	RMSE	R^2
Stacked RNN	Many-to-One	1.5200	4.3343	2.0819	0.9308
Stacked LSTM	Many-to-One	1.5585	4.4213	2.1027	0.9294
Stacked GRU	**Many-to-One**	**1.5267**	**4.2160**	**2.0533**	**0.9327**

Interpretation. The stacked GRU had the lowest RMSE (2.0533), lowest MSE (4.2160), and highest R^2 (0.9327) for the matched Many-to-One task. The stacked RNN had a marginally lower MAE (1.5200 compared with the GRU's 1.5267); therefore the GRU did not win every metric.

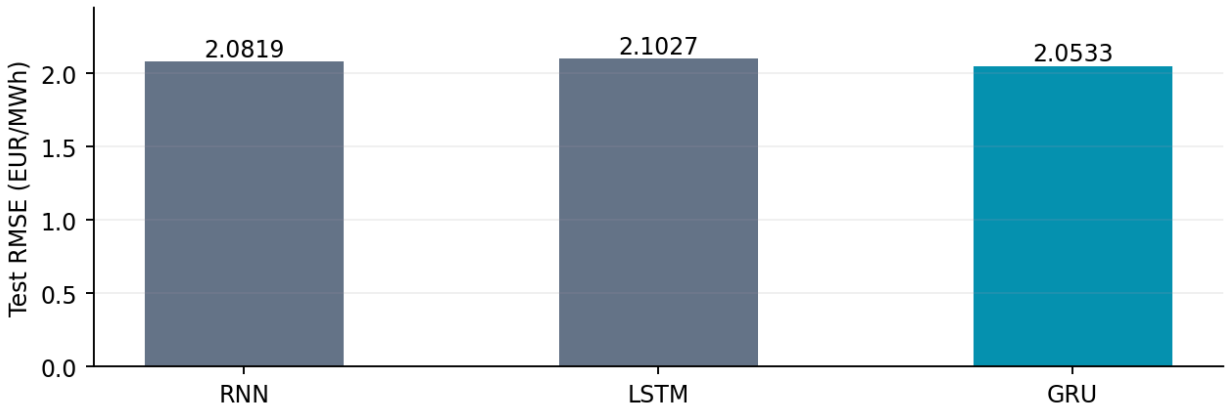


Figure 1. Many-to-One Comparison among the Three Stacked Recurrent Families

The synchronized Many-to-Many models also produced relatively strong results: LSTM RMSE 2.1855 and GRU RMSE 2.2015. One-to-Many performed considerably worse (RMSE 7.0506–7.7728, R^2 0.0337–0.2049), consistent with the greater difficulty of predicting six future values from one observed timestep in this experiment.

Comparisons between different typologies are descriptive, not strictly like-for-like, because their input histories, forecast horizons, and output alignments differ. The most defensible model-family comparisons are within the same typology and test partition. Single-run differences should not be interpreted as statistically significant without repeated seeds or uncertainty estimates.

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4. Advanced Challenge: Add One CNN Layer Before the RNN for 100 epochs

One one-dimensional convolutional layer was inserted immediately before the SimpleRNN stage in each of the five original RNN typologies. The layer used 32 filters, kernel size 3, and ReLU activation. All five CNN–RNN models were trained for 100 epochs with batch size 64 and Adam optimization.

```
Conv1D(filters=32, kernel_size=3,
    padding='causal' if typology == 'many_to_many_synced' else 'same',
    activation='relu')

# Followed by the original RNN recurrent blocks and forecast head.
model.fit(X_train, y_train, validation_data=(X_val, y_val),
      epochs=100, batch_size=64)
```

The synchronized Many-to-Many experiment uses causal padding. An earlier implementation using the same padding allowed the kernel to read the next input timestep—including the value being predicted—creating target leakage. The incorrect result was archived, and only the affected CNN–RNN model was retrained from scratch for 100 epochs. The four other typologies retain the same padding because their forecast targets follow the observed input sequence.

Measured CNN–RNN Test Performance

Typology	MAE	MSE	RMSE	R^2
One-to-One	2.0225	8.0060	2.8295	0.8718
Many-to-One	1.5334	4.3645	2.0891	0.9303
One-to-Many	6.0053	57.4078	7.5768	0.0818
Many-to-Many (Synced)	1.6536	5.0356	2.2440	0.9196
Many-to-Many (Seq2Seq)	3.6678	22.4721	4.7405	0.6415

5. Conclusion on the CNN–RNN Model with 32 Filters and Kernel Size 3

The 100-epoch CNN–RNN achieved mixed outcomes across sequence typologies. Its lowest RMSE was observed for Many-to-One (2.0891, R^2 0.9303), while synchronized Many-to-Many achieved RMSE 2.2440 and R^2 0.9196 after correction for future-target leakage. In contrast, One-to-Many produced RMSE 7.5768 and R^2 0.0818, reflecting weak predictive fit in this experimental configuration.

The convolutional layer can learn local patterns from the 24-hour input windows before recurrent processing, but extra features and parameters do not ensure improved forecasting. In One-to-One and One-to-Many, the input consists of just one timestep, so a width-3 kernel has no genuine three-step temporal neighborhood to extract. Thus, CNN usefulness depends strongly on the input/output structure.

Overall, using Conv1D with 32 filters and kernel size 3 was beneficial for selected window-based forecasting tasks, but it did not consistently outperform the original RNN. The corrected results support a targeted rather than universal benefit from CNN feature extraction.

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6. Compare CNN–RNN Performance Against the Original RNN for Every Typology

The original configurations provided the baseline for each corresponding CNN–RNN experiment. Both variants used the same dataset partitions, training settings, and forecasting targets. For source One-to-Many and Seq2Seq, "original one-layer RNN" refers to the original one-layer encoder; their retained decoder is a second recurrent block.

Paired MAE, MSE, RMSE and R^2 Results

Typology	Architecture	MAE	MSE	RMSE	R^2
One-to-One	Original RNN	2.0140	7.9997	2.8284	0.8719
	CNN–RNN	2.0225	8.0060	2.8295	0.8718
Many-to-One	Original RNN	1.5590	4.6478	2.1559	0.9258
	CNN–RNN	1.5334	4.3645	2.0891	0.9303
One-to-Many	Original RNN	5.6087	54.1231	7.3568	0.1344
	CNN–RNN	6.0053	57.4078	7.5768	0.0818
Many-to-Many (Synced)	Original RNN	1.8096	5.7520	2.3983	0.9082
	CNN–RNN	1.6536	5.0356	2.2440	0.9196
Many-to-Many (Seq2Seq)	Original RNN	3.4684	20.8115	4.5620	0.6680
	CNN–RNN	3.6678	22.4721	4.7405	0.6415

RMSE Change Relative to the Original RNN

Typology	Original RMSE	CNN RMSE	Change	Outcome
One-to-One	2.8284	2.8295	+0.04%	Worsened
Many-to-One	**2.1559**	**2.0891**	**-3.10%**	**Improved**
One-to-Many	7.3568	7.5768	+2.99%	Worsened
Many-to-Many (Synced)	**2.3983**	**2.2440**	**-6.43%**	**Improved**
Many-to-Many (Seq2Seq)	4.5620	4.7405	+3.91%	Worsened

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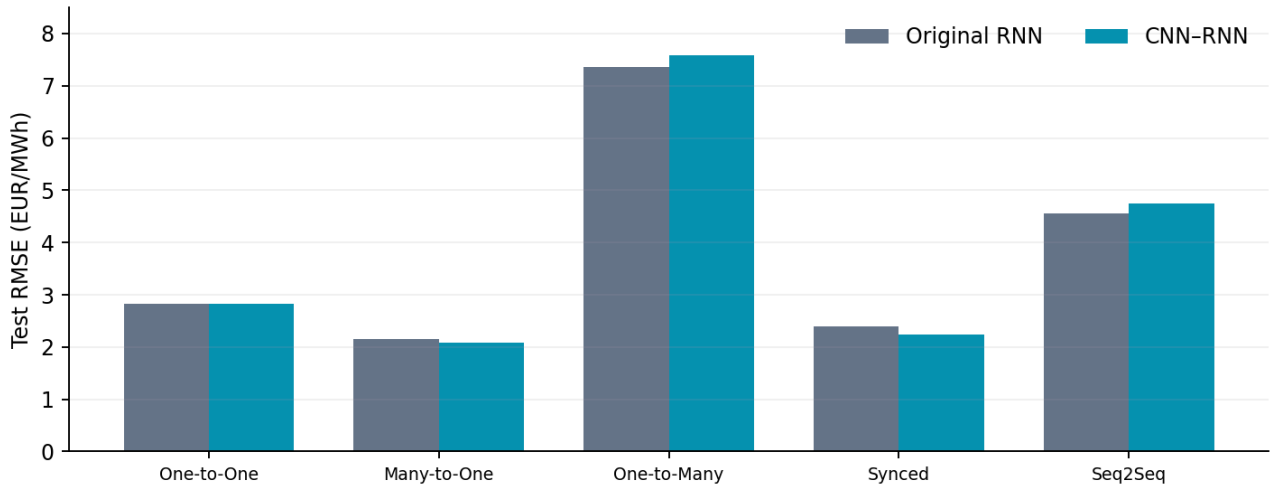


Figure 2. Original RNN and CNN–RNN Test RMSE for Each Matched Typology

One-to-One: CNN–RNN was essentially unchanged: RMSE rose from 2.8284 to 2.8295 (+0.04%). A one-timestep input limits temporal pattern extraction by a kernel of width 3.

Many-to-One: CNN–RNN reduced RMSE from 2.1559 to 2.0891 (−3.10%) and improved R^2 from 0.9258 to 0.9303. The convolution plausibly helped identify local patterns in the 24-hour history before recurrent processing.

One-to-Many: RMSE increased from 7.3568 to 7.5768 (+2.99%), and R^2 dropped from 0.1344 to 0.0818. The one-timestep input provides insufficient temporal context for meaningful local convolutions across three real timesteps.

Many-to-Many Synchronized: With causal convolution, RMSE decreased from 2.3983 to 2.2440 (−6.43%), and R^2 increased from 0.9082 to 0.9196. This was the largest valid relative RMSE improvement, but it should not be confused with the superseded noncausal result.

Many-to-Many Seq2Seq: RMSE rose from 4.5620 to 4.7405 (+3.91%), while R^2 fell from 0.6680 to 0.6415. Adding convolution did not improve this multi-step encoder–decoder forecasting task in the measured run.

Final Comparison: CNN–RNN improved two of five matched typologies (Many-to-One and synchronized Many-to-Many), with deterioration or negligible change in the other three. More elaborate architectures do not inherently outperform simpler recurrent networks; accuracy depends on the particular temporal task and the information supplied to the model. This evaluation is restricted to one dataset, one chronological split, and the recorded training runs.